DI WATER CLEANING

PCBA Assembly Process Guide

Deionized Water Systems • Wash Equipment • Process Parameters
Quality Control • Residue Analysis • Environmental Standards

PCBA Manufacturing & Assembly Standards
IPC-A-610 | IPC/J-STD-001 | RoHS | MIL-STD-202

1. OVERVIEW OF DI WATER CLEANING IN PCBA

Deionized (DI) water cleaning is a critical process step in PCBA manufacturing that removes flux residues, solder paste contaminants, ionic residues, and particulates from printed circuit board assemblies after soldering. Proper cleaning ensures long-term reliability, prevents electrochemical migration, and meets customer cleanliness requirements for high-reliability applications (Tesla, Zoox, medical, military).

Why DI Water Cleaning is Essential:

- **Removes flux residues:** Active flux residues are hygroscopic and corrosive
- **Prevents dendritic growth:** Ionic contamination causes electrochemical migration
- **Ensures conformal coating adhesion:** Clean surface required for coating bonding
- **Improves electrical performance:** Reduces leakage currents and impedance issues
- **Meets military/medical standards:** IPC-A-610 Class 3 and MIL-STD requirements
- **RoHS compliance:** Water-soluble lead-free fluxes require mandatory cleaning

■ **Key Principle:** The effectiveness of DI water cleaning depends on four factors: **water purity** (resistivity), **temperature** (60–70°C optimal), **mechanical action** (spray pressure, ultrasonics), and **drying method** (hot air, vacuum, or IR).

2. DEIONIZED WATER SYSTEM

2.1 What is Deionized Water?

Deionized (DI) water is water that has had its mineral ions (cations and anions) removed through an ion exchange process. Unlike distilled water (which removes contaminants via evaporation), DI water specifically targets ionic content, making it ideal for electronics cleaning.

DI Water Specifications for PCBA Cleaning:

- **Resistivity:** ≥ 1 MΩ·cm (megohm-centimeter) — 18 MΩ·cm ideal
- **Conductivity:** ≤ 1 µS/cm (microsiemens per centimeter)
- **pH:** 6.5–7.5 (neutral)
- **Total Dissolved Solids (TDS):** < 1 ppm
- **Particles:** 0.2 µm filtration or better
- **Organic content:** < 0.1 ppm TOC

2.2 DI Water Generation System

A typical in-house DI water system for PCBA cleaning consists of multiple stages:

Stage 1: Pre-Treatment

- **Sediment filter:** Removes particulates > 5 µm
- **Carbon filter:** Removes chlorine, organic compounds, and odors
- **Water softener:** Removes calcium and magnesium (hardness ions)

Stage 2: Reverse Osmosis (RO)

- **RO membrane:** Removes 95–99% of dissolved salts and contaminants
- **RO water quality:** ~10–50 µS/cm conductivity
- **Rejection rate:** 98% of dissolved solids removed

Stage 3: Deionization (DI)

- **Mixed-bed ion exchange resin:** Cation and anion resins in single vessel
- **Final quality:** 1–18 MΩ·cm resistivity
- **Resin regeneration:** Required when resistivity drops below 1 MΩ·cm

Stage 4: Polishing & Storage

- **UV sterilization:** Kills bacteria in storage tank
- **0.2 µm final filter:** Removes any resin fines or particles
- **Storage tank:** Polyethylene or stainless steel, recirculating loop

■ **Best Practice:** Monitor DI water resistivity continuously with an inline meter. Regenerate or replace mixed-bed resin when resistivity falls below 1 MΩ·cm. Use 18 MΩ·cm water for final rinse of high-reliability assemblies.

3. DI WATER CLEANING EQUIPMENT

3.1 Batch Wash Systems (Ultrasonic + DI Water)

Batch wash systems are the most common in-house cleaning method for low-to-medium volume PCBA production. Boards are loaded into a basket and submerged in a heated DI water bath with ultrasonic agitation.

System Components:

- **Ultrasonic generator:** 40 kHz typical (28 kHz for heavy contamination)
- **Heated tank:** 60–70°C DI water
- **Spray under immersion (SUI):** Recirculating spray nozzles below liquid level
- **Rinse tank:** Second tank with fresh DI water
- **Dry chamber:** Hot air knife or vacuum drying

Process Cycle:

1. **Wash:** 5–10 minutes in heated DI water with ultrasonics
2. **Rinse:** 2–5 minutes in fresh DI water (no ultrasonics)
3. **Final Rinse:** 1–2 minutes in high-purity DI water (18 MΩ·cm)
4. **Dry:** 10–20 minutes hot air at 80–100°C

3.2 Inline Conveyor Wash Systems

Inline systems are used for high-volume production where boards pass continuously through multiple cleaning zones. These systems integrate directly into the SMT line after reflow.

System Zones:

- **Pre-wash zone:** Heated DI water spray (60–70°C) to soften residues
- **Main wash zone:** High-pressure spray (2–5 bar) with rotating nozzles
- **Rinse zone 1:** Fresh DI water spray to remove wash water
- **Rinse zone 2:** High-purity DI water (18 MΩ·cm) final rinse
- **Dry zone:** Hot air knives + IR heaters (100–120°C)

Conveyor Speed:

- **Typical:** 0.5–2.0 meters per minute
- **Adjust based on:** Board complexity, flux type, contamination level

3.3 Spray-in-Air vs. Ultrasonic Comparison

Feature	Spray-in-Air	Ultrasonic
Cleaning Action	High-pressure spray jets	Cavitation bubbles

Best For	Low-profile, open boards	Dense, complex, under-component
Component Risk	Low — gentle mechanical action	Medium — potential damage to delicate parts
Under-component Cleaning	Moderate	Excellent
Throughput	High (inline)	Low-Medium (batch)
Water Consumption	Higher	Lower
Cost	Higher initial investment	Lower initial investment

Table 1: Spray-in-Air vs. Ultrasonic Cleaning Comparison

4. DI WATER CLEANING PROCESS PARAMETERS

4.1 Water Temperature

Temperature is the most critical parameter for effective DI water cleaning. Higher temperatures reduce water surface tension, improve solubility of flux residues, and accelerate chemical reactions.

- **Wash Temperature:** 60–70°C (140–158°F)
- **Rinse Temperature:** 60–70°C (same as wash)
- **Final Rinse:** Room temperature to 40°C (prevents water spots)
- **Max Temperature:** 80°C (risk of component damage or water spots)

4.2 Spray Pressure

Spray pressure determines mechanical cleaning force. Too low = insufficient residue removal. Too high = component displacement or damage.

- **Pre-wash:** 1–2 bar (gentle softening)
- **Main wash:** 2–5 bar (active cleaning)
- **Rinse:** 1–3 bar (removal of wash water)
- **Final rinse:** 0.5–1 bar (gentle, no spots)

4.3 Wash Time

Wash time depends on flux type, board density, and contamination level.

- **Water-soluble flux (WS-820, WS-809):** 5–10 minutes
- **RMA flux residues:** 10–15 minutes
- **No-clean flux (if cleaning required):** 15–20 minutes
- **Heavy contamination / rework:** 15–25 minutes

4.4 Drying Parameters

Incomplete drying leaves moisture trapped under components, causing corrosion and electrical failures.

- **Hot air temperature:** 80–120°C
- **Dry time:** 10–30 minutes depending on board thickness and density
- **Air velocity:** 5–15 m/s
- **IR drying:** Optional for thick boards or high-density assemblies
- **Vacuum drying:** Optional for ultra-high reliability (removes trapped moisture)

■■ **Critical:** Verify complete drying before packaging. Use a moisture indicator card or weigh boards before and after drying. Any weight gain indicates trapped moisture.

5. FLUX TYPE & CLEANING REQUIREMENTS

Different flux chemistries have different cleaning requirements. Using the wrong cleaning process for a given flux type results in incomplete residue removal or board damage.

Flux Type	IPC Code	Cleaning Required	DI Water Effective	Notes
Water-Soluble (WS)	ORH1, WSM1	■ Mandatory	■ Excellent	Residues hygroscopic & corrosive
RMA (Rosin Mildly Active)	ROL1, REM1	■■ Optional	■■ Moderate	May need saponifier additive
No-Clean (Low Residue)	REL0, ORL0	■ Not Required	■ Not Effective	Residues designed to remain benign
No-Clean (High Residue)	REL1, ORH1	■■ Sometimes	■■ Difficult	May require semi-aqueous cleaner
RA (Rosin Activated)	ROH1, REH1	■ Mandatory	■ Poor	Requires solvent or saponifier
Alpha WS-820	ORH0/1	■ Mandatory	■ Excellent	In-house primary LF paste
Alpha OM-353	REL0/1	■ Not Required	N/A	In-house no-clean LF paste
Alpha WS-809	WSM1	■ Mandatory	■ Excellent	In-house SnPb washable paste
AIM WS715M	ORM0/1	■ Mandatory	■ Excellent	In-house water-soluble flux
Alpha RF-800	ROL0/1	■ Not Required	N/A	In-house no-clean flux

Table 2: In-House Flux Types & DI Water Cleaning Requirements

■ **In-House Rule:** All water-soluble paste and flux production (WS-820, WS-809, WS715M) **MUST** pass through DI water cleaning. No-clean materials (OM-353, RF-800) bypass cleaning unless customer specifically requires residue removal.

6. QUALITY CONTROL & CLEANLINESS TESTING

6.1 Ionic Contamination Testing (ROSE Method)

The **Resistivity of Solvent Extract (ROSE)** test measures total ionic contamination on a cleaned PCBA. It is the industry standard for quantifying cleaning effectiveness.

Test Method (IPC-TM-650 2.3.25):

1. Extract board in 75% IPA / 25% DI water solution
2. Measure solution resistivity before and after extraction
3. Calculate equivalent NaCl contamination in µg/cm²

Acceptance Limits:

- **IPC-A-610 Class 1 (General):** ≤ 1.56 µg/cm² NaCl equivalent
- **IPC-A-610 Class 2 (Dedicated Service):** ≤ 1.56 µg/cm²
- **IPC-A-610 Class 3 (High Performance):** ≤ 0.78 µg/cm² (or ≤ 1.56 µg/cm² per customer)
- **MIL-STD-202 Method 310:** ≤ 1.56 µg/cm²
- **Medical / Implantable:** ≤ 0.65 µg/cm² (strictest requirement)

6.2 Surface Insulation Resistance (SIR)

SIR testing measures electrical resistance between closely spaced conductors on the PCBA surface. Low SIR indicates ionic contamination that can cause leakage currents or dendritic growth.

- **Test Conditions:** 85°C / 85% RH for 7 days (or 40°C / 90% RH for 21 days)
- **Minimum SIR:** ≥ 10■ Ω (100 MΩ) for acceptable cleanliness
- **Test Pattern:** IPC-B-25A standard comb pattern or customer-specific

6.3 Visual Inspection

Visual inspection under magnification (5–40×) checks for visible residues, water spots, and component damage after cleaning.

- **Acceptable:** No visible white residues, no water spots, no component displacement
- **Reject:** White powdery residues (flux residue or mineral deposits)
- **Reject:** Water spots indicating incomplete drying
- **Reject:** Discolored or damaged components

6.4 Ion Chromatography (IC)

Ion Chromatography identifies specific ionic species (F■, Cl■, Br■, SO■²■, etc.) and their concentrations. Used for root-cause analysis when ROSE fails.

- **Detection limit:** 0.01 ppm for most ions
- **Sample preparation:** Extract in DI water, inject into IC column
- **Turnaround:** 1–2 days (lab analysis)

■ **In-House Testing Schedule:** Perform ROSE testing on 1 board per lot for Class 2 production. Perform ROSE + SIR on 100% of Class 3 (medical/military) lots. Document all results in the quality management system.

7. TROUBLESHOOTING COMMON CLEANING ISSUES

7.1 White Residue After Cleaning

White residue is the most common cleaning defect. It can be flux residue, mineral deposits from poor-quality water, or corrosion products.

Causes & Solutions:

- **Cause:** Insufficient wash time or temperature

→ **Solution:** Increase wash time to 10–15 min; raise temperature to 65–70°C

- **Cause:** DI water resistivity too low

→ **Solution:** Regenerate mixed-bed resin; verify ≥ 1 MΩ·cm

- **Cause:** No-clean flux being cleaned (residue not water-soluble)

→ **Solution:** Use semi-aqueous or solvent cleaner; or switch to washable flux

- **Cause:** Hard water deposits (calcium/magnesium)

→ **Solution:** Verify RO system operation; check water softener

7.2 Water Spots

Water spots are mineral deposits left after evaporation. They indicate either poor water quality or incomplete drying.

- **Cause:** Final rinse water not pure enough

→ **Solution:** Use 18 MΩ·cm water for final rinse

- **Cause:** Insufficient drying time or temperature

→ **Solution:** Extend dry cycle; increase air temperature to 100–120°C

- **Cause:** Slow conveyor speed in inline dryer

→ **Solution:** Reduce conveyor speed or add IR heating zone

7.3 Component Damage

Delicate components can be damaged by excessive ultrasonic energy or high spray pressure.

- **MEMS microphones:** Disable ultrasonics; use spray-only cleaning
- **Crystal oscillators:** Reduce ultrasonic frequency to 28 kHz or lower
- **LEDs:** Reduce wash temperature to < 60°C
- **RF shields:** Verify no water trapping under shields; add drain holes in design

7.4 Incomplete Drying

Moisture trapped under large components (BGA, QFN, connectors) causes long-term reliability issues.

- **Solution:** Extend dry time to 30+ minutes for dense boards
- **Solution:** Use vacuum drying for high-reliability assemblies
- **Solution:** Design boards with drain holes under large components
- **Solution:** Angle boards during drying to promote drainage

8. ENVIRONMENTAL & SAFETY CONSIDERATIONS

8.1 Water Conservation

DI water cleaning consumes significant water. Implement closed-loop systems where possible:

- **Recirculating wash:** Filter and reuse wash water (with flux load monitoring)
- **Counter-flow rinse:** Fresh water enters final rinse, cascades to earlier stages
- **RO reject recovery:** Use RO concentrate for non-critical cleaning (floors, tools)
- **Conductivity monitoring:** Auto-dump wash water when flux load exceeds threshold

8.2 Waste Water Treatment

Wash water containing flux residues may require treatment before discharge:

- **pH adjustment:** Neutralize acidic or alkaline waste streams
- **Heavy metals:** Test for lead, tin, silver; treat if above discharge limits
- **Evaporation:** Reduce volume via evaporator; dispose concentrate as hazardous waste
- **Contract disposal:** Use certified waste management for concentrated flux waste

8.3 Operator Safety

DI water cleaning systems present several safety hazards:

- **Hot surfaces:** Wash tanks and dryers operate at 60–120°C — burn hazard
- **Slip hazard:** Wet floors around cleaning equipment
- **Electrical:** Ensure all equipment is properly grounded; GFCI protection
- **Chemical:** Flux residues in wash water may be irritants — wear gloves and eye protection

■■ **Safety Rule:** Never open hot wash chambers or dryers during operation. Allow equipment to cool to < 40°C before maintenance. Lockout/tagout procedures mandatory for all cleaning equipment servicing.

9. IN-HOUSE DI WATER CLEANING PROCESS FLOW

This section maps the complete DI water cleaning workflow for in-house PCBA production, from reflow exit to final inspection.

Step 1: Reflow Exit & Cool Down

- Boards exit reflow oven at ~100°C
- Allow natural cooling to < 60°C before cleaning (prevents thermal shock)
- Visual check for obvious defects before cleaning

Step 2: Load into Cleaning System

- Place boards in stainless steel basket (batch) or on conveyor (inline)
- Ensure boards are not stacked — water must reach all surfaces
- Secure loose components with fixtures if needed

Step 3: DI Water Wash

- Fill tank with 60–70°C DI water (≥ 1 MΩ·cm)
- Activate ultrasonics (40 kHz) or spray system (2–5 bar)
- Wash duration: 5–15 minutes depending on flux type
- Monitor water conductivity; replace when > 50 µS/cm

Step 4: DI Water Rinse

- Transfer to fresh DI water rinse tank
- Rinse duration: 2–5 minutes
- Agitation: Gentle spray or manual brush for under-component areas

Step 5: Final High-Purity Rinse

- Rinse with 18 MΩ·cm DI water for 1–2 minutes
- This step removes any residual ions from previous stages
- Critical for Class 3 (medical/military) assemblies

Step 6: Drying

- Hot air drying at 80–120°C for 10–30 minutes
- Verify complete dryness (no water spots, no moisture under components)
- Optional: Vacuum drying for ultra-high reliability

Step 7: Quality Verification

- Visual inspection under 10× magnification
- ROSE test (1 board per lot for Class 2; 100% for Class 3)
- Record results in quality management system

Step 8: Packaging or Next Process

- Boards must be completely dry before conformal coating
- Use moisture-barrier bags if storage > 24 hours before next step
- Label with cleaning date, operator, and test results

■ **In-House Tip:** For WS-820 (water-soluble SAC305 paste) production, complete the entire cleaning cycle within 2 hours of reflow exit. Delayed cleaning allows flux residues to harden, making removal more difficult.

10. QUICK REFERENCE SUMMARY

DI Water Specifications:

→ **Resistivity:** ≥ 1 MΩ·cm (18 MΩ·cm for final rinse)

→ **Temperature:** 60–70°C (wash and rinse)

→ **pH:** 6.5–7.5 (neutral)

→ **TDS:** < 1 ppm

Cleaning Parameters:

→ **Wash time:** 5–15 minutes (flux-dependent)

→ **Spray pressure:** 2–5 bar (main wash)

→ **Ultrasonic:** 40 kHz (standard); 28 kHz (heavy contamination)

→ **Dry time:** 10–30 minutes at 80–120°C

Quality Limits:

→ **ROSE (Class 2):** ≤ 1.56 µg/cm^2 NaCl equivalent

→ **ROSE (Class 3):** ≤ 0.78 µg/cm^2 NaCl equivalent

→ **SIR:** ≥ 10■ Ω after 85°C/85% RH for 7 days

→ **Visual:** No white residues, no water spots, no damage

In-House Materials Requiring Cleaning:

→ **ALPHA WS-820** — Water-soluble SAC305 paste (MANDATORY wash)

→ **ALPHA WS-809** — Water-soluble SnPb paste (MANDATORY wash)

→ **AIM WS715M** — Water-soluble liquid flux (MANDATORY wash)

→ **ALPHA OM-353** — No-clean (bypass unless customer requires)

→ **ALPHA RF-800** — No-clean (bypass unless customer requires)

Troubleshooting Quick Fixes:

→ **White residue:** Increase wash time; check DI water resistivity

→ **Water spots:** Use 18 MΩ·cm final rinse; extend drying

→ **Incomplete drying:** Add vacuum dry step; angle boards

→ **Component damage:** Reduce ultrasonic power; use spray-only

References & Standards:

• IPC-A-610: Acceptability of Electronic Assemblies
• IPC/J-STD-001: Requirements for Soldered Electrical and Electronic Assemblies
• IPC-TM-650 2.3.25: Resistivity of Solvent Extract (ROSE) Test
• IPC-TM-650 2.6.3.3: Surface Insulation Resistance (SIR) Test

• MIL-STD-202 Method 310: Contamination Testing
• ZESTRON Technical Cleaning Guides
• ALPHA & AIM Product Technical Datasheets

End of Document — DI Water Cleaning PCBA Assembly Complete Guide